A.S. 2024-2025

Investment Strategies

Flexible Asset Allocation

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Introduction

This paper focuses on understanding the logic underlying Flexible Asset Allocation (FAA). The analysis conducted was developed taking into account the work published on the Alpha Architect website, which in turn can be traced back to the paper published by Keller and Putten "Generalized Momentum and Flexible Asset Allocation (FAA): A Heuristic Approach".

The FAA strategy reproduced on Alpha Architect integrates **momentum**, **volatility** and **correlation** into the determination of the risk regime, and adjusts weights between **7 different assets**. The approach we chose was different in terms of the choice of assets and in terms of the number of the initial basket of securities that made up the available securities universe.

Some backtests on the FAA strategy **are conducted in the study**. The FAA classifies assets based on three factors:

- **momentum**,
- **volatility**,
- **correlation**.

The best strategy discussed in the original paper assigns:

- a **100% weight to the 4-month relative momentum**,
- a **50% weight to volatility over 4 months**,
- a **50% weight to the correlation over 4 months**.

In particular, the approach proposed by Wesley Gray focuses on the choice of 3 assets chosen from a universe of 7 assets.

The purpose of the following paper is to evaluate the persistence and effectiveness of these strategies on a broader universe of securities and a more stringent selection in asset allocation. The initial study was an attempt to verify whether the results reported by the authors were also confirmed on a different sample and with a different universe of assets.

An alternative approach is also proposed, in which changes are made in terms of weighting and the measure of volatility, without however distorting it in its original underlying logic.

Dataset

As anticipated, it was decided to use a larger dataset to test the actual validity of the selection methods proposed by the strategy on a wider universe of securities. It was therefore decided to opt for a dataset that is as complete and exhaustive as possible in terms of asset class. It can be divided into three macro-areas:

- **Bonds**: for the bond sector, both government and corporate instruments were considered, with different maturities in order to obtain greater completeness.
  More recently introduced, inflation-linked instruments have also been introduced, for which the time series of prices are shorter, given their creation a posteriori compared to the initial date of the rest of the dataset. This approach allows for backtesting that takes into account financial innovations that have been widely used by investors since their introduction into the markets.
- **Equity**: for the equity sector, completeness in geographical terms was sought in order to have a basket of securities that included developed, emerging and global markets. It was decided not to introduce tools built on Smart Beta logics into the dataset, both for the relatively recent introduction of these tools, and to have a starting dataset that was faithful to the one originally used by the authors of the experiment.
- **Commodity**: for the commodity sector, only one representative asset has been chosen.
  The SPGSCITR Index is in fact one of the leading global commodity indices and measures the performance of a large and diversified basket of physical commodities. The index has been chosen for its comprehensiveness It includes over 20 commodities traded on the world's main futures markets, divided into 5 major sectors:
  - Energy
  - Industrial metals
  - Precious metals
  - Agricultural products
  - Livestock

The selected assets are detailed below, each included in its reference category:

- **Short Term Govies:**
  - W1G1 Index – Bloomberg U.S. Treasury Bill 1-3 Month Index
  - W1GE Index – Bloomberg U.S. Treasury Bill Index
- **Govies**

  - W0G1 Index – Bloomberg Global Treasury Bill 1–3 Month Index
  - W0GE Index – Bloomberg Global Treasury Bill Index
  - G0Q0 Index – Bloomberg Global Aggregate Treasury Total Return Index
- **Inflation Linked Bonds**
  - W0GI Index – Bloomberg Global Treasury Total Return Index
  - EG0I Index – Bloomberg Euro Treasury Total Return Index
  - G0QI Index – Bloomberg Global Aggregate Treasury Index
- **Investment Grade Bonds**
  - G0BC Index – Bloomberg Global Aggregate Corporate Total Return Index
  - C0A0 Index – Bloomberg U.S. Corporate Total Return Index
  - ER00 Index – Bloomberg Euro Aggregate Treasury Total Return Index
- **High Yield Bond**
  - HW00 Index – Bloomberg Global High Yield Corporate Total Return Index
  - HE00 Index – Bloomberg Euro High Yield Corporate Total Return Index
  - H0A0 Index – Bloomberg U.S. Corporate High Yield Total Return Index
- **Emerging Market Bond**
  - JPEIGLBL Index – JPM Emerging Markets Bond Index (EMBI) Global / Diversified / Local
- **Global Equities**
  - MXWD Index – MSCI All Country World Index (ACWI)
- **Developed Equities**
  - MXWO Index – MSCI World Index
  - SPX Index – S&P 500 Index
  - SXXP Index – STOXX Europe 600 Price Index
  - DAX Index – German Stock Index (DAX)
  - CAC Index – CAC 40 Index
  - UKX Index – FTSE 100 Index
  - TPX Index – TOPIX Index (Tokyo Stock Price Index)
- **Emerging Markets Equities**
  - MXEF Index – MSCI Emerging Markets Index
- **Commodity Index**
  - SPGSCITR Index – S&P GSCI Commodity Total Return Index

The choice of **the risk-free** asset fell on **the "W1G1 Index – Bloomberg U.S. Treasury Bill 1-3 Month Index".** The choice to use a US based risk-free was made to ensure continuity with the original paper, in which "VFISX" was used. An asset was therefore chosen that can be considered as a good proxy for the original choice of the authors.

Performance Metrics

In order to comprehensively evaluate the effectiveness and robustness of the investment strategies under analysis, the function developed in Matlab calculates an articulated set of performance, risk and co-movement metrics with the benchmark. The measures implemented include traditional indicators, widely used in financial literature and professional practice, as well as advanced statistics aimed at providing a more granular assessment of the risk-return profile.

1. Traditional return and risk metrics

- **CAGR (Compound Annual Growth Rate):** measures the average compound annual growth of the portfolio, highlighting the strategy's ability to generate value over the long term.
- **Annualized Return and Annualized Volatility**: indicate the average annualized return and annual volatility, respectively, providing a summary assessment of the risk-return trade-off.
- **Sharpe Ratio**: assesses the return in excess of the risk-free rate per unit of risk taken, allowing a standardized comparison between strategies with different levels of risk.

2. Advanced downside and risk-adjusted metrics

- **Sortino Ratio**: improves the interpretation of the risk profile by focusing only on negative volatility compared to a minimum acceptable return (MAR), in our analysis set at 0.
- **Modified Sortino Ratio**: Uses a composite measure of negative risk to provide a more robust indication of the quality of returns versus losses.
- **Worst Drawdown and Calmar Ratio**: measure the maximum cumulative loss compared to previous highs and the ratio between return and maximum drawdown, assessing the resilience of the portfolio during adverse market phases.

3. Robustness and stability metrics

- **Validity Index**: combines cumulative return and drawdown characteristics (frequency and duration) to offer a synthetic indicator of the robustness of the strategy over time.

- **Best/Worst Month and Profitable Months**: provide an overview of the distribution of returns, highlighting the frequency and intensity of extreme events and the consistency of performance.

- **Rolling Win Percentage (1Y, 5Y, 10Y):** measures the probability that the strategy will generate positive returns over medium-long term rolling horizons, assessing the stability of performance.

4. Extreme Risk Metrics

- **Value at Risk (VaR) and Expected Shortfall (ES):** estimate the potential loss at a given confidence level and the average loss conditioned in worst-case scenarios, respectively, providing a measure of tail risk.

5. Co-movement metrics with the benchmark

- **Up/Down Capture Ratio and Conditional Correlations**: assess the strategy's ability to outperform the benchmark in bull and bear markets.

- **Tracking Error**: measures the standard deviation of the return in excess of the benchmark, indicative of the strategy's active risk.

Portfolio Rating

In order to have a synthetic, universal and intuitive metric for comparing different strategies and portfolios, a proprietary indicator called **Omega Portfolio Rating (OPR)** has been developed.The objective of the OPR is to express, in percentage terms, the improvement or worsening of a strategy compared to a reference benchmark – in this case, the Equally Weighted portfolio with monthly rebalancing – through a weighted combination of three independent measures of efficiency and robustness of the investment.

In particular, the OPR represents **the average of the percentage change of three specific metrics**, selected to offer an overall assessment of the strategy's risk-adjusted profile:

1. **Calmar Ratio** – a consolidated measure that expresses the ratio of the Compound Annual Growth Rate (CAGR) to the maximum drawdown; it assesses the efficiency of the return adjusted for the maximum sustained loss.
2. **Modified Sortino Ratio** – variant of the traditional Sortino ratio, in which the denominator is not the standard deviation of negative returns, but the arithmetic mean of negative returns. This change emphasizes the penalty due to the frequency and intensity of average losses, rather than their variability.

3. **Validity Index** – a new ad hoc indicator, designed to measure the solidity and consistency of performance over time. In the numerator, it considers the cumulative return of the strategy excluding the top 1% of returns, so as to reduce the impact of positive and isolated extreme events. The denominator is instead the average between the maximum duration (in terms of time) of drawdowns and the average duration of the overall drawdowns observed, thus providing a measure of risk in time dimension.

$$OPR = \frac{1}{3}\left(\frac{S_p - S_b}{|S_b|} + \frac{C_p - C_b}{|C_b|} + \frac{R_p - R_b}{|R_b|}\right)$$

Taken together, these three components make it possible to assess in an integrated way the strategy's ability to generate sustainable and regular returns, while limiting exposure to prolonged drawdowns. The final result of the Omega Portfolio Rating therefore provides a synthetic index (centered on zero) that allows you to appreciate, in an immediate and comparable way, how much a given strategy exceeds or underperforms the benchmark in terms of overall efficiency and resilience of the risk/return profile.

Original strategy

This section takes a closer look at the original strategy and implementation method proposed by Alpha Architect for its replication.

The basic idea of Flexible Asset Allocation is based on the desire to extend the traditional logic of momentum strategies to additional parameters to be considered when defining the asset allocation at the time of each monthly rebalance.

The implementation of the different Backtests was approached gradually, based on the rules established by each of the strategies devised. Below are the 4 types of strategies proposed on Alpha Architect and adapted for the approach we have chosen:

- **A – Relative momentum:** Assets are sorted by 4-month **relative momentum**; you invest in the 7 highest ranked assets.
- **RA – Relative momentum + Absolute momentum:** assets are sorted by relative momentum; **if the absolute momentum is greater than 1**, you invest, **otherwise you switch to the risk-free asset**.
- **RAV – Relative momentum + Absolute momentum + Volatility:** sorting by relative momentum and volatility. The final score is given by **100% relative momentum + 50% volatility**. If the absolute momentum is > 1, you invest; otherwise, you switch to risk-free.

- **RAVC – Relative momentum + Absolute momentum + Volatility + Correlation:** sorting by relative momentum, volatility, and **average correlation**. The overall score is: **100% relative momentum + 50% volatility + 50% correlation**. If the absolute momentum is > 1, you invest; otherwise, we switch to risk-free.

In addition, we use the following legend:

- **EW_BM – Equal-weight benchmark:** $w_i = \frac{1}{N}$
  This is the equal-weight benchmark portfolio, rebalanced monthly based on the number of assets available. The visible differences in terms of allocation depend exclusively on the late start of the time series of some specific types of assets of more recent introduction (e.g. inflation-linked bonds).
- **MA 1-12 – Moving average filter:**
  The Moving Average 1-12 strategy (also known as **the dual moving average**) compares the short-term moving average (1 month) to the long-term moving average (12 months) to generate investment signals. The purpose of the same is to generate a signal based on prices:
  Signal = $\begin{cases} 1 \; se \; MA_1(t) \; > \; MA_{12}(t) \\ 0 \; se \; MA_1(t) \; \leq \; MA_{12}(t) \end{cases}$
  - $MA_1(t)$= 1-month moving average (last month's price)
  - $MA_{12}(t)$= 12-month moving average
    Signal = 1 → investment in the asset
    Signal = 0 → risk-free

In the faithful replication of the strategy, **the same logic of the classic strategy has been re-proposed, with changes in terms of the choice of assets adapted on the basis of the dataset we use.** It is therefore specified that the only change made in this section consists of a more stringent selection of assets in which to invest, 7 out of 25 available, compared to what was implemented in the strategy proposed by the authors of the paper (3 out of 7).

It was also decided to include the Moving Average strategy in the Backtest, to remain faithful to the original purpose of the experiment: "Flexible Asset Allocation: Dethroning Moving Average Rules?".

The fundamental approach is to assign a score to each asset based on the lookback period that has been set. The methodologies and the consequent construction of the different scores can be seen gradually:

- **Relative Momentum (R):** $\boldsymbol{L_i = w_R \cdot \text{rank}(r_i)}$
  The score awarded builds a simple ranking among the assets. This allows you to identify the top assets to which to assign the default weight of the momentum on returns of 1 and therefore the score. An equally weighted asset allocation is envisaged on the selected top assets.

- **Absolute Momentum (RA):** $\boldsymbol{L_i = w_R \cdot \text{rank}(r_i)}$
  The same approach is used for absolute momentum, but in a more stringent way. In fact, only top assets with positive momentum are considered. In the event that there are no 7 assets that meet this requirement, the uninvested part is allocated in risk-free.

- **Generalized Momentum (RAV):** $\boldsymbol{L_i = w_R \cdot \text{rank}(r_i) + w_V \cdot \text{rank}(v_i)}$
  To include **volatility and other factors** in the equation, the authors introduced the **L "loss function"** for the portfolio, where a higher value of L indicates a **worse risk/return profile**.

$$L = L(r, v)$$

where and are the vectors of the returns and volatility of our assets, with the following derivatives:$r\ v\ r_i\ v_i U$

- $\frac{\partial L}{\partial r_i} < 0$: Higher performance is betterr_i
- $\frac{\partial L}{\partial v_i} > 0$: lower volatility is betterv_i

It simply allows you **to order different portfolios or funds**, where a lower value **of L** always represents a **better risk/return combination**.

Rank() = 1$\boldsymbol{v_i}$ (best) is attributed when the volatility of the fund is i **the lowest** among all 7 funds in the universe, and **rank() = 7**$\boldsymbol{v_i}$ (worst) when it is **the highest**.

It can therefore be considered as a L_i **generalized ranking function**, since it is simply a linear function of the rankings of the two factors (return) and (volatility), in which rank = 1 is always the best.

The weights and determine the relative importance of the high return ranking is relative to the volatility ranking in our $w_R w_V$ **generalized momentum function**. Both weights must be **non-negative**.

Once you have chosen and , you can $w_R w_V$ **sort all the funds by** $L_i$and select the **best** Nones (e.g., 3 out of 7). In practice, **the ranking based on** r_i(as done in the

relative momentum) has been replaced by the **ranking based on** L_i, the generalized measure.

- **Generalized Momentum (RAVC):** $\boldsymbol{L_i = w_R \cdot \text{rank}(r_i) + w_V \cdot \text{rank}(v_i) + w_C \cdot \text{rank}(c_i)}$
  The FAA's last stage takes correlation into account as an additional factor. Portfolio diversification improves when the average correlation between assets is lower, i.e. when the non-diagonal elements of the correlation matrix are smaller.
  The generalized ranking function (normalized through rankings) can be easily extended to include correlation.c_i
  It is therefore understandable that even in this case, a lower correlation should be rewarded in the ranking, in order to encourage greater diversification of the portfolio.
  For the latter category of strategies, different strategies with different lookback periods have been implemented, in continuity with what was presented in the final section of the Alpha Architect project.

The analysis carried out was developed starting from logarithmic returns such as , for the following reasons:$r_t = \ln\,(P_t/P_{t-1})$

1. **Time additivity**: over a compound horizon, the total logarithmic return is the sum of the returns of the individual periods, simplifying time aggregations.

2. **Compatibility with statistical models**: log-returns often show properties closer to the assumptions of normality and linearity required by AR models, GARCH, regressions, etc.

3. **Continuous growth**: Log-return corresponds to a continuous growth rate, consistent with exponential growth patterns.

4. **Consistency with respect to different horizons**: allows direct comparison between horizons (daily, monthly, yearly) by adding the log-returns.

Strategy replication

In the paper, the data period runs from **January 1, 2005 to December 31, 2012.** The data period chosen by Alpha Architect has been extended to the most recent date available for analysis, July **31, 2014**.

The breadth and depth of the dataset available to us enabled us to analyse different strategies over a longer period, allowing us to assess their resilience during several

different types of market cycles. The Dataset available to us goes from **31 December 1987 to 30 June 2017**. It was decided to adopt a subdivision of the analysis into two sub-periods in addition to the analysis on the entire time series. The 3 periods analysed at the operational level were:

- **IN SAMPLE: from December 31, 1987 and December 31, 2006**
- **OUT OF SAMPLE**: **from December 31, 2006 to June 30, 2017**
- **FULL SAMPLE: from December 31, 1987 to June 30, 2017**.

The subdivision of the dataset was carried out in continuity with the choice of the authors of the paper and its replication. Among the **decisive reasons** linked to this choice, there are the desire to:

- **Avoid overestimating performance** (a typical problem with overfitting).
- Allow you **to evaluate whether the model works in contexts other than those used to calibrate it**.
- Provide a **more realistic measure of operational effectiveness**.

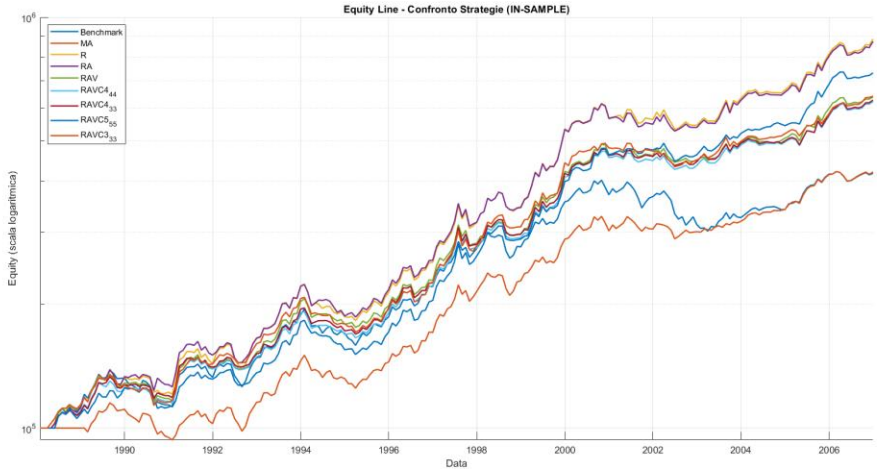
The division of the dataset between **In-Sample (60%)** and **Out-of-Sample (40%)** responds to the need to rigorously evaluate the effectiveness and robustness of the strategy. The **In-Sample** range, equal to 60% of the total period, is large enough to allow model calibration and parameter optimization on a statistically significant representative sample of historical data. At the same time, it leaves an **Out-of-Sample** portion large enough to test the model's ability to generalize on data never used in the construction phase. This distinction reduces the risk of *overfitting* and allows you to verify whether the performance observed during development remains stable even under different market conditions.

As with the original strategy, **the strategies were evaluated without taking into account transaction costs**. This promise is necessary if we consider the monthly rebalancing that would be carried out in the event of real-life implementation of the strategies analyzed. The decision not to include transaction costs stems from the desire to carry out a mere comparison between the different strategies, to measure the effectiveness introduced by the founding elements of the individual strategies. Improvements may include the introduction of transaction costs to verify the effectiveness of strategies in an investment environment.

ANALYSIS IN SAMPLE

The first part of the analysis of the strategies implemented focuses on the In Sample period. It was decided to identify the period **between 31 December 1987 and 31 December 2006** as the period of time over which to test the strategies on a preliminary basis.

Equity Lines



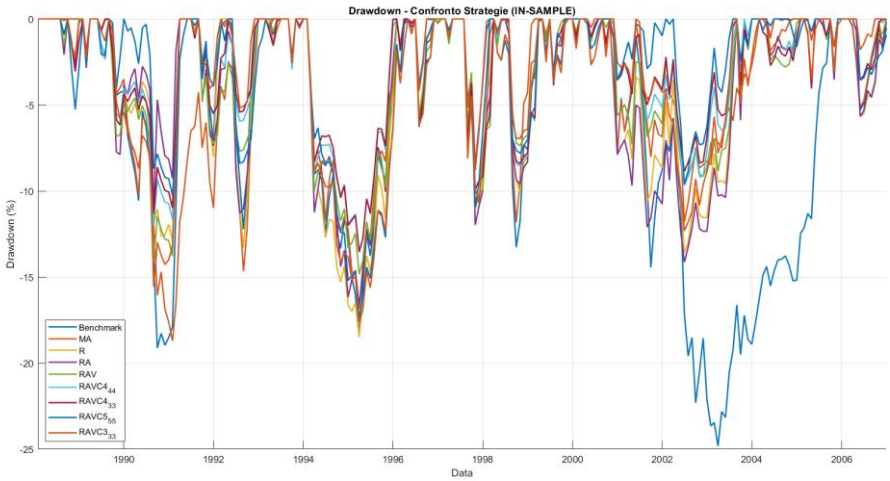
Looking only at the equity comparison chart between the strategies, interesting signals can be seen that will be confirmed in all the graphic trends of the backtest. In particular, three subgroups within the strategies can be noted quite clearly:

- The **moving average strategy and the equally weighted benchmark have performed quite similarly**.
- The **Relative and Absolute momentum strategies show a rather similar trend**, with slight deviations in the months characterized by the absence of assets in which to invest in the Absolute strategy.
- **Multi-factor strategies have shown similar performance**. The positive impact on the riskiness of these strategies given by the introduction of volatility and correlation factors is already graphically visible.

These first impressions are confirmed by the further analysis carried out on this backtest period.

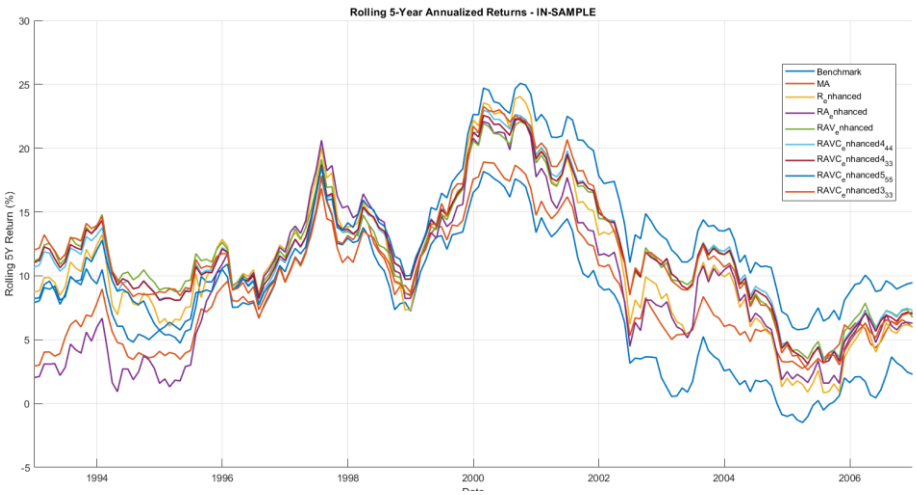
Moreover, the equity curves suggest that the advanced strategies achieved smoother growth trajectories and smaller drawdowns compared to the simpler models. This indicates that **incorporating additional risk-based factors enhances portfolio stability without significantly compromising returns**. The consistency of these results across multiple time horizons further reinforces the robustness of the multi-factor framework.

Drawdowns



Greater resilience to drawdowns is observed in strategies that integrate risk analysis, achieved through the use of volatility and correlation measures.

Rolling 5-year annualized returns



Correlation matrix

	Benchmark	MA	R	RA	RAV	RAVC4_4_4	RAVC4_3_3	RAVC5_5_5	RAVC3_3_3
Benchmark	1	0,89	0,85	0,8	0,79	0,78	0,77	0,79	0,78
MA	0,89	1	0,84	0,79	0,78	0,78	0,77	0,81	0,79
R	0,85	0,84	1	0,98	0,96	0,95	0,95	0,93	0,91
RA	0,8	0,79	0,98	1	0,97	0,96	0,96	0,92	0,91
RAV	0,79	0,78	0,96	0,97	1	0,97	0,97	0,94	0,92
RAVC4_4_4	0,78	0,78	0,95	0,96	0,97	1	0,99	0,94	0,94
RAVC4_3_3	0,77	0,77	0,95	0,96	0,97	0,99	1	0,94	0,94
RAVC5_5_5	0,79	0,81	0,93	0,92	0,94	0,94	0,94	1	0,9
RAVC3_3_3	0,78	0,79	0,91	0,91	0,92	0,94	0,94	0,9	1

Summary statistics

Strategy	CAGR	Volatility	Sharpe	Sortino	SortinoMod	Calmar	Validity	Omega
Benchmark	0,0784	0,1038	0,249	0,4051	0,3025	0,316	0,0674	0
MA	0,0788	0,0919	0,2852	0,4577	0,3498	0,4211	0,1047	0,6735
R	0,1222	0,1201	0,5465	0,8756	0,4329	0,6604	0,2788	2,5664
RA	0,1214	0,1194	0,5436	0,8652	0,4322	0,729	0,2728	2,5715
RAV	0,103	0,1032	0,4693	0,7668	0,4363	0,6945	0,194	2,2664
RAVC4_4_4	0,1016	0,0974	0,4833	0,7994	0,4666	0,7493	0,1937	2,5386
RAVC4_3_3	0,102	0,0979	0,4856	0,8028	0,4607	0,7532	0,1947	2,536
RAVC5_5_5	0,111	0,0978	0,5685	0,8726	0,4719	0,6363	0,2941	2,6951
RAVC3_3_3	0,1033	0,0967	0,5031	0,7937	0,4529	0,5851	0,2463	2,2335

Strategy	MaxDD	BestMonth	WorstMonth	ProfitableMonths	Roll1YWin	Roll5YWin	Roll10YWin
Benchmark	-0,2482	0,092	-0,0819	0,6035	0,7083	0,9167	1
MA	-0,1872	0,0788	-0,0722	0,5683	0,7407	1	1
R	-0,185	0,1132	-0,0946	0,5947	0,787	1	1
RA	-0,1665	0,1132	-0,0946	0,5947	0,8009	1	1
RAV	-0,1484	0,1079	-0,0861	0,6079	0,7824	1	1
RAVC4_4_4	-0,1355	0,119	-0,0771	0,6035	0,7685	1	1
RAVC4_3_3	-0,1355	0,119	-0,0771	0,6167	0,7685	1	1
RAVC5_5_5	-0,1744	0,0977	-0,0835	0,63	0,8102	1	1
RAVC3_3_3	-0,1766	0,1091	-0,0846	0,63	0,7824	1	1

Correlation analysis

Strategy	UpPct	DownPct	NegCorrPct	PosCorrPct	TrackingErrorPct
Benchmark	100%	100%	100%	100%	0%
MA	88,32%	87,36%	77,37%	74,93%	4,68%
R	86,13%	80,46%	59,81%	76,83%	6,28%
RA	83,94%	77,01%	46,28%	72,74%	7,22%
RAV	83,94%	73,56%	47,29%	71,89%	6,66%
RAVC4_4_4	82,48%	72,41%	48,13%	68,73%	6,77%
RAVC4_3_3	82,48%	68,97%	47,64%	69,68%	6,82%
RAVC5_5_5	85,40%	70,11%	57,65%	65,42%	6,50%
RAVC3_3_3	85,40%	71,26%	48,59%	69,64%	6,65%

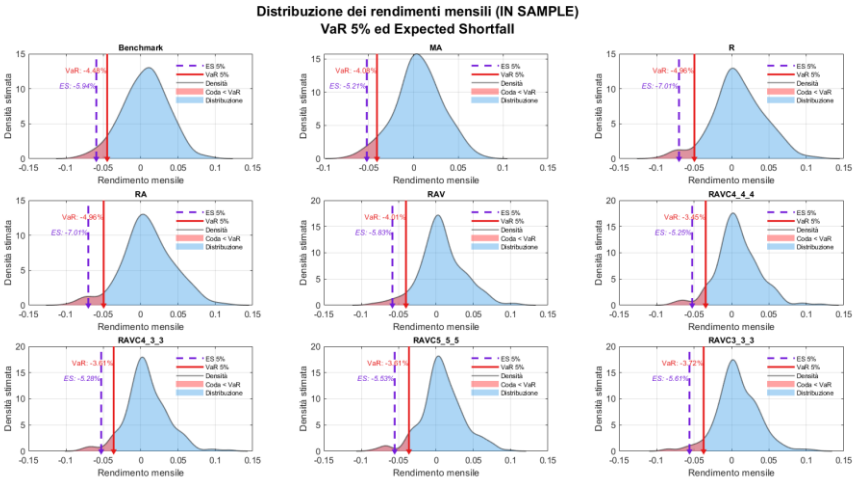
Tail events analysis

An additional analysis we carry out includes **the assessment of the VaR at 95% and the expected shortfall at the same confidence level**. These metrics were added to our overall assessment as we felt we **needed to focus more on possible tail events**. The representation of the monthly return distributions of each strategy was approximated with a continuous distribution, despite the discrete nature of the observations.

In the case of financial time series, particularly in the case of monthly returns of an investment strategy, the observed data are **decent**. You have a finite number of empirical observations (e.g., monthly returns recorded in the backtest period). To analyze the statistical properties of these returns in more depth – in particular to study the shape of the distribution, estimate tail risk and calculate measures such as **Value at Risk (VaR)** and **Expected Shortfall (ES)** – it is common and useful **to approximate the observed discrete distribution with an estimated continuous distribution**. This approximation has been achieved through **probability density estimation techniques**, in particular kernel *density estimation* (KDE). In this way, a continuous function is obtained that interpolates the discrete points of the observed returns and that allows you to:

- **Graphically represent** the distribution in a more informative and readable way than a simple histogram;
- **Estimate cumulative and quantile probabilities** with greater precision, facilitating the calculation of risk measures such as VaR, which is based on distribution-specific percentiles;
- **Analyze deployment queues** (where extreme events are located) more accurately, improving the risk assessment of rare but significant losses.

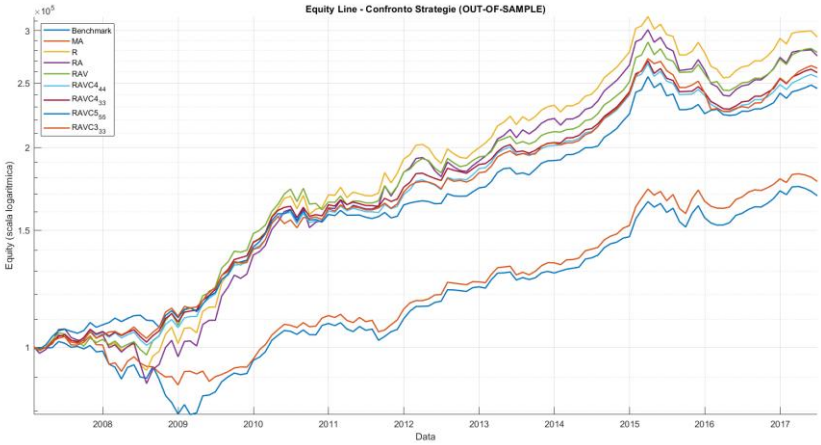
Strategy	VaR5pct	ES5pct
Benchmark	-0,0448	-0,0594
MA	-0,0408	-0,0521
R	-0,0496	-0,0701
RA	-0,0496	-0,0701
RAV	-0,0401	-0,0583
RAVC4_4_4	-0,0345	-0,0525
RAVC4_3_3	-0,0361	-0,0528
RAVC5_5_5	-0,0361	-0,0553
RAVC3_3_3	-0,0372	-0,0561



OUT OF SAMPLE ANALYSIS

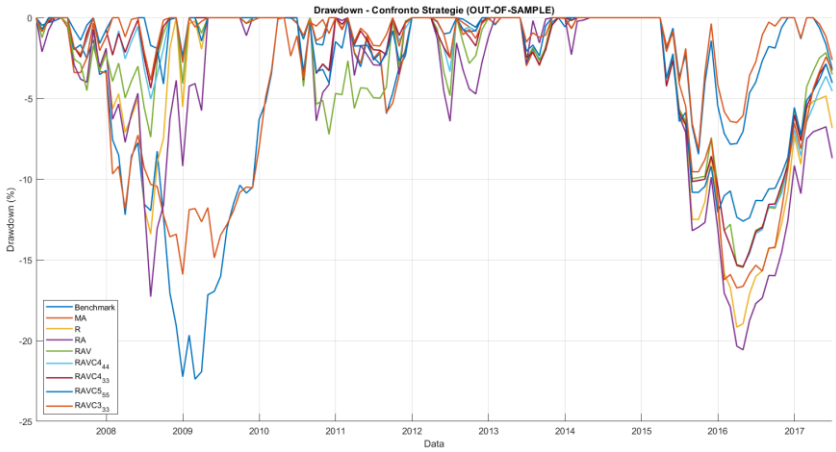
For the Out Of Sample analysis, data **from 1 January 2007 to 30 June 2017** were taken into consideration. This allowed us to use the remaining 40% of the dataset for the actual analysis of strategies. **Including the 2008 financial crisis in the out-of-sample dataset was crucial to be able to test the robustness and real predictive capacity of investment strategies**. A period of extreme stress such as that of 2008 represents a highly non-linear and volatile environment, different from ordinary market phases. Evaluating the performance of strategies in this scenario allows us to verify how resilient they are to sudden shocks, avoiding overly optimistic conclusions based only on periods of growth and thus providing a more realistic and comprehensive assessment of their effectiveness.

Equity lines

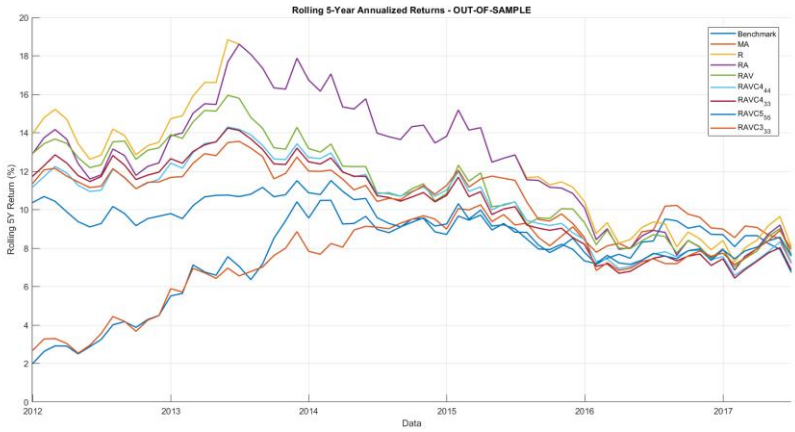


Advanced strategies confirm their superiority over the benchmark and the moving average-based strategy.

Drawdowns



Rolling 5-year annualized returns



Correlation matrix

Strategy	Benchmark	MA	R	RA	RAV	RAVC4_4_4	RAVC4_3_3	RAVC5_5_5	RAVC3_3_3
Benchmark	1	0,86	0,66	0,65	0,63	0,64	0,64	0,59	0,57
MA	0,86	1	0,65	0,65	0,62	0,66	0,67	0,64	0,57
R	0,66	0,65	1	0,99	0,96	0,93	0,93	0,81	0,84
RA	0,65	0,65	0,99	1	0,95	0,93	0,93	0,81	0,83
RAV	0,63	0,62	0,96	0,95	1	0,94	0,94	0,86	0,85
RAVC4_4_4	0,64	0,66	0,93	0,93	0,94	1	0,99	0,89	0,91
RAVC4_3_3	0,64	0,67	0,93	0,93	0,94	0,99	1	0,89	0,91
RAVC5_5_5	0,59	0,64	0,81	0,81	0,86	0,89	0,89	1	0,84
RAVC3_3_3	0,57	0,57	0,84	0,83	0,85	0,91	0,91	0,84	1

Summary statistics

Strategy	CAGR	Volatility	Sharpe	Sortino	Sortino Mod	Calmar	Validity	Omega
Benchmark	0,0526	0,0763	0,2587	0,0429	0,2521	0,2351	0,0286	0
MA	0,0578	0,0707	0,3488	0,1416	0,316	0,3635	0,0324	0,8433
R	0,1101	0,1034	0,7051	0,8093	0,4019	0,5746	0,0968	2,8338
RA	0,1032	0,1066	0,6252	0,6691	0,3531	0,5017	0,0849	2,1904
RAV	0,104	0,0862	0,7822	0,9553	0,4977	0,6723	0,0916	3,5685
RAVC4_4_4	0,0944	0,0689	0,8525	0,9428	0,5544	0,6129	0,083	3,4404
RAVC4_3_3	0,096	0,0682	0,8817	0,9911	0,5806	0,6231	0,0861	3,6233
RAVC5_5_5	0,0903	0,0669	0,8216	0,8089	0,5039	0,7169	0,0771	3,6134
RAVC3_3_3	0,0977	0,0695	0,8866	0,9661	0,5627	0,5837	0,0915	3,4483

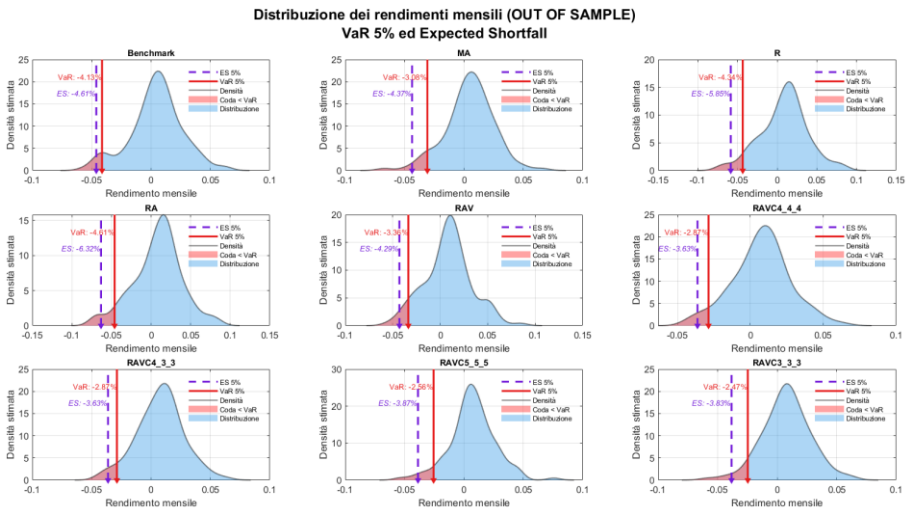
Strategy	MaxDD	BestMonth	WorstMonth	ProfitableMonths	Roll1YWin	Roll5YWin	Roll10YWin
Benchmark	-0,2239	0,0626	-0,0559	0,6508	0,7217	1	1
MA	-0,1591	0,0657	-0,0671	0,619	0,7478	1	1
R	-0,1916	0,0844	-0,0732	0,6746	0,8261	1	1
RA	-0,2057	0,0844	-0,0732	0,6746	0,7913	1	1
RAV	-0,1547	0,0845	-0,055	0,6746	0,8174	1	1
RAVC4_4_4	-0,1541	0,0631	-0,0434	0,6984	0,8696	1	1
RAVC4_3_3	-0,1541	0,0631	-0,0434	0,6905	0,87	1	1
RAVC5_5_5	-0,126	0,0756	-0,0541	0,7222	0,8609	1	1
RAVC3_3_3	-0,1673	0,0631	-0,0582	0,6905	0,9043	1	1

Correlation analysis

Strategy	UpPct	DownPct	NegCorrPct	PosCorrPct	TrackingErrorPct
Benchmark	100%	100%	100%	100%	0%
MA	87,80%	86,36%	72,50%	72,18%	3,88%
R	86,59%	68,1.818%	21,94%	59,52%	7,78%
RA	85,37%	65,91%	19%	58,95%	8,10%
RAV	84,15%	63,64%	9,62%	58,08%	7,09%
RAVC4_4_4	86,59%	61,36%	15,79%	56,72%	6,16%
RAVC4_3_3	85,37%	61,36%	12,86%	56,31%	6,54%
RAVC5_5_5	85,37%	50%	17,96%	50,37%	6,50%
RAVC3_3_3	84,15%	56,82%	8,60%	49,80%	6,78%

Tail events analysis

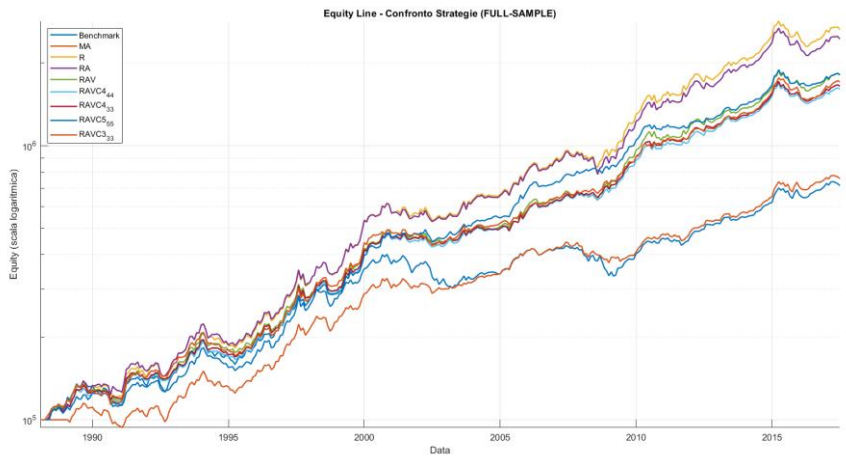
Strategy	VaR5pct	ES5pct
Benchmark	-0,0413	-0,0461
MA	-0,0308	-0,0437
R	-0,0434	-0,0585
RA	-0,0461	-0,0632
RAV	-0,0336	-0,0429
RAVC4_4_4	-0,0287	-0,0363
RAVC4_3_3	-0,0287	-0,0363
RAVC5_5_5	-0,0256	-0,0387
RAVC3_3_3	-0,0247	-0,0383



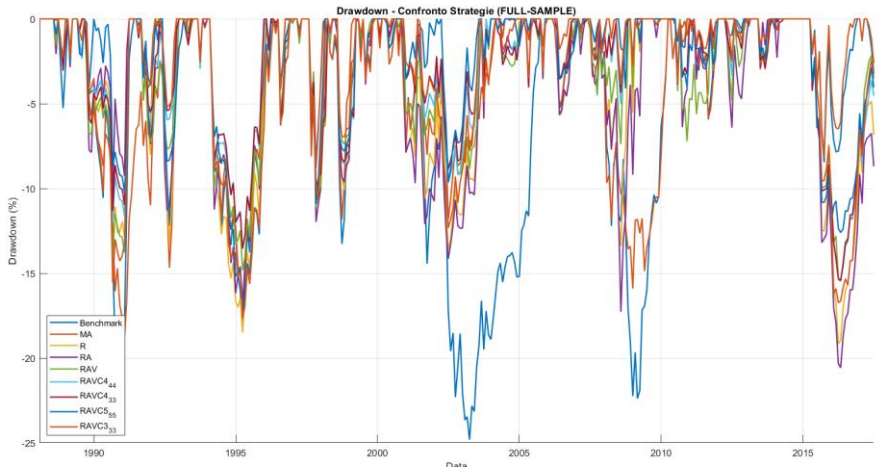
The Out Of Sample backtest offers the possibility to assess **the actual persistence of the benefits brought by the introduction of volatility and correlations as elements capable of influencing the asset allocation**.

FULL SAMPLE ANALYSIS

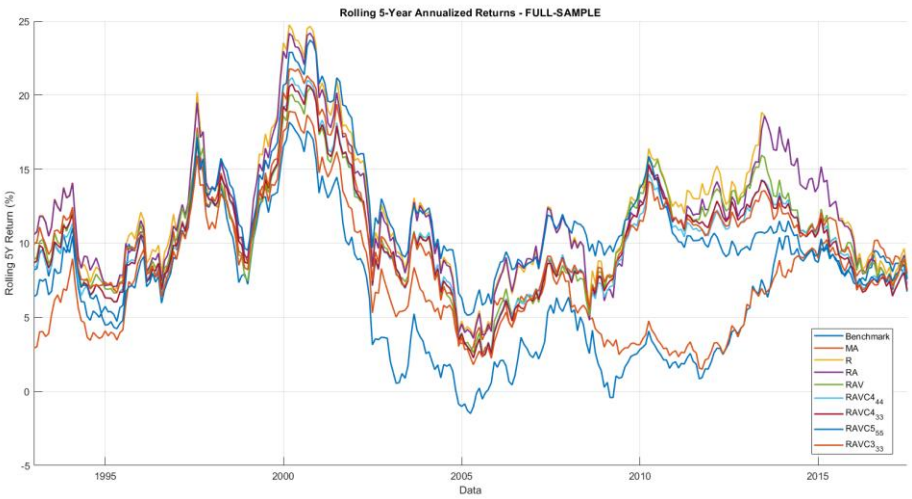
Equity lines



Drawdowns



Rolling 5-year annualized returns

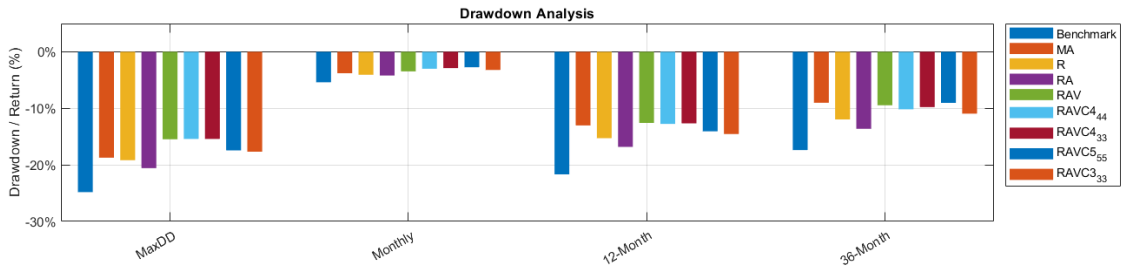


Correlation matrix

	Benchmark	MA	R	RA	RAV	RAVC4_4_4	RAVC4_3_3	RAVC5_5_5	RAVC3_3_3
Benchmark	1	0,89	0,8	0,76	0,75	0,75	0,74	0,75	0,73
MA	0,89	1	0,79	0,75	0,74	0,75	0,75	0,77	0,74
R	0,8	0,79	1	0,98	0,96	0,94	0,94	0,89	0,89
RA	0,76	0,75	0,98	1	0,96	0,95	0,95	0,89	0,89
RAV	0,75	0,74	0,96	0,96	1	0,96	0,96	0,93	0,93
RAVC4_4_4	0,75	0,75	0,94	0,95	0,96	1	0,99	0,93	0,93
RAVC4_3_3	0,74	0,75	0,94	0,95	0,96	0,99	1	0,93	0,93
RAVC5_5_5	0,75	0,77	0,89	0,89	0,93	0,93	0,93	1	0,89
RAVC3_3_3	0,73	0,74	0,89	0,89	0,93	0,93	0,93	0,89	1

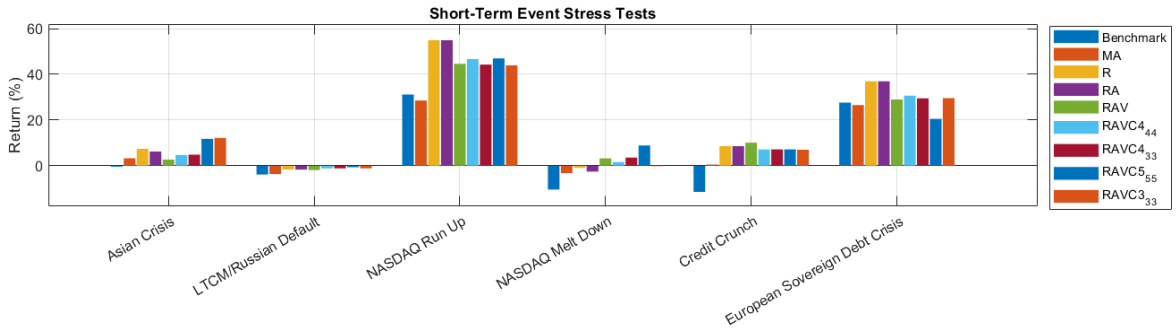
For the Full Sample backtest, additional evaluations have been introduced compared to what was done for the other two periods analyzed. This methodology was developed in line with what was presented by the authors of the replication of the strategy on Alpha Architect.

Drawdown analysis



An ad hoc analysis was carried out on the drawdowns of the different strategies. Also in this case, **the most advanced strategies are particularly effective in mitigating risk**.

Short-term event stress tests



The analysis of short-term events highlights the **major trend following trend of classic momentum strategies, both absolute and relative**. As expected**, these strategies are more closely involved in short-term positive return events, while being more affected by negative ones**. More advanced strategies prove to be excellent at mitigating the risks

of the most basic strategies, while being able to participate in positive trends, although this happens more conservatively than R and RA.

Annual returns of individual strategies

Anno	Benchmark	MA	R	RA	RAV	RAVC4_4_4	RAVC4_3_3	RAVC5_5_5	RAVC3_3_3
1988	13,39%	0,00%	11,89%	11,89%	13,76%	9,91%	9,26%	8,25%	13,97%
1989	16,65%	11,11%	18,76%	19,33%	13,16%	13,67%	16,64%	15,93%	11,55%
1990	-14,97%	-14,63%	-8,01%	-4,50%	-8,32%	-7,01%	-5,93%	-9,29%	-9,08%
1991	18,15%	8,04%	17,41%	18,07%	18,62%	19,90%	17,39%	15,56%	23,82%
1992	11,67%	12,74%	15,54%	9,91%	9,30%	7,13%	7,65%	3,69%	12,18%
1993	27,44%	26,17%	33,81%	33,81%	32,83%	27,86%	28,61%	32,88%	28,66%
1994	-10,05%	-9,47%	-15,86%	-15,37%	-11,90%	-11,43%	-11,52%	-14,41%	-13,18%
1995	9,76%	9,77%	14,69%	16,06%	10,90%	13,02%	13,62%	10,48%	8,55%
1996	18,33%	17,98%	18,34%	17,87%	15,23%	16,37%	15,35%	19,29%	15,86%
1997	23,98%	25,90%	25,10%	23,71%	23,08%	23,56%	23,83%	28,36%	23,25%
1998	0,76%	6,11%	8,91%	8,54%	5,12%	5,86%	6,01%	9,93%	11,11%
1999	28,65%	26,16%	54,94%	54,94%	42,44%	40,30%	40,56%	38,00%	43,29%
2000	4,10%	4,79%	6,26%	6,26%	9,80%	11,61%	10,55%	16,22%	8,61%
2001	-1,00%	1,08%	-0,85%	-3,12%	-0,50%	-1,65%	-0,60%	2,67%	-1,74%
2002	-14,63%	-1,51%	-3,14%	-1,76%	-2,63%	-2,66%	-2,82%	-2,59%	-4,82%
2003	4,13%	6,07%	16,43%	16,43%	10,57%	12,14%	10,84%	12,67%	10,86%
2004	4,60%	6,84%	4,66%	4,66%	2,40%	1,72%	1,28%	4,44%	4,47%
2005	21,92%	21,58%	22,40%	22,40%	18,16%	15,67%	15,79%	26,18%	11,85%
2006	0,53%	1,43%	8,71%	8,55%	6,51%	8,38%	8,56%	6,52%	10,26%
2007	-0,04%	2,27%	6,46%	6,66%	4,39%	6,26%	6,70%	9,21%	5,27%
2008	-19,61%	-12,94%	-2,92%	-7,44%	5,03%	1,77%	3,41%	2,48%	6,85%
2009	20,49%	9,36%	42,32%	42,32%	37,42%	31,33%	31,75%	27,82%	26,18%
2010	13,64%	16,31%	17,73%	17,73%	11,58%	14,11%	13,91%	11,97%	15,04%
2011	1,61%	2,26%	13,17%	13,17%	10,95%	5,88%	6,27%	3,33%	5,97%
2012	11,75%	9,94%	4,03%	4,03%	5,42%	9,45%	8,25%	6,11%	6,98%
2013	4,92%	6,83%	15,90%	15,90%	9,18%	8,34%	7,89%	9,97%	10,88%
2014	13,42%	13,81%	20,35%	20,35%	18,81%	19,18%	19,24%	17,49%	18,68%
2015	6,82%	8,80%	-0,27%	-1,71%	2,85%	-0,18%	-0,23%	0,14%	0,28%
2016	9,78%	8,08%	5,08%	4,49%	5,32%	3,86%	4,91%	7,38%	5,28%
2017	-1,70%	-0,88%	0,50%	0,50%	2,27%	2,59%	2,12%	1,61%	3,47%

Summary statistics

Strategy	CAGR	Volatility	Sharpe	Sortino	SortinoMod	Calmar	Validity	Omega
Benchmark	0,0691	0,0948	0,2496	0,2848	0,2844	0,2786	0,1119	0
MA	0,0713	0,0849	0,3023	0,3525	0,338	0,3808	0,1862	0,7767
R	0,1179	0,1143	0,5971	0,8516	0,4217	0,615	0,7306	3,5337
RA	0,1148	0,1149	0,5705	0,7971	0,4038	0,5584	0,6636	3,0681
RAV	0,1034	0,0973	0,5672	0,816	0,4539	0,6683	0,5146	3,1948
RAVC4_4_4	0,099	0,0882	0,5809	0,8241	0,4845	0,6426	0,4773	3,9099
RAVC4_3_3	0,0999	0,0883	0,5891	0,8384	0,4868	0,6481	0,488	3,1586
RAVC5_5_5	0,1035	0,0879	0,6295	0,8363	0,472	0,5939	0,6611	3,4278
RAVC3_3_3	0,1013	0,0879	0,6064	0,8278	0,4776	0,5737	0,608	3,2168

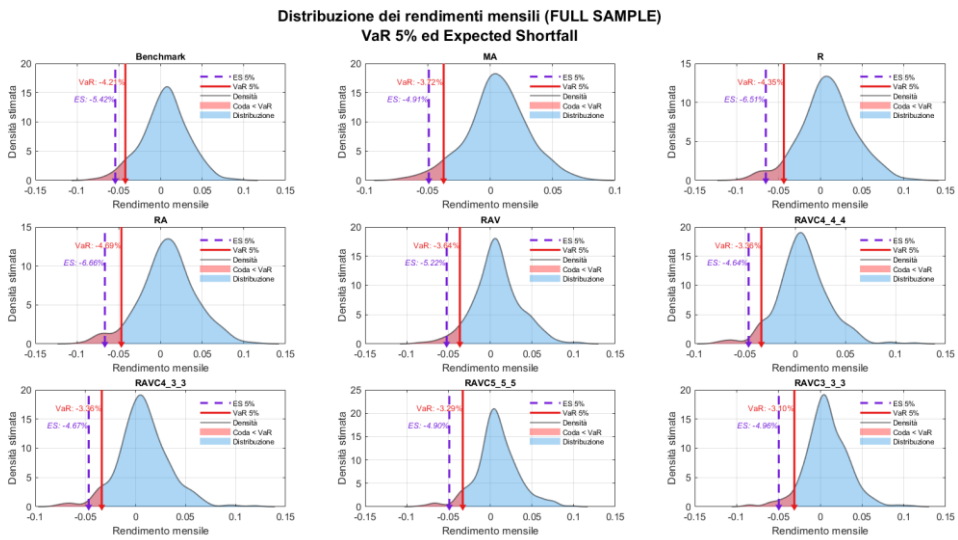
Strategy	MaxDD	BestMonth	WorstMonth	ProfitableMonths	Roll1YWin	Roll5YWin	Roll10YWin
Benchmark	-0,2482	0,092	-0,0819	0,6204	0,7193	0,9422	1
MA	-0,1872	0,0788	-0,0722	0,5864	0,7515	1	1
R	-0,1916	0,1132	-0,0946	0,6232	0,807	1	1
RA	-0,2057	0,1132	-0,0946	0,6232	0,8041	1	1
RAV	-0,1547	0,1079	-0,0861	0,6317	0,8012	1	1
RAVC4_4_4	-0,1541	0,119	-0,0771	0,6374	0,8099	1	1
RAVC4_3_3	-0,1541	0,119	-0,0771	0,6431	0,8158	1	1
RAVC5_5_5	-0,1744	0,0977	-0,0835	0,6629	0,8304	1	1
RAVC3_3_3	-0,1766	0,1091	-0,0846	0,6516	0,8304	1	1

Correlation analysis

Strategy	UpPct	DownPct	NegCorrPct	PosCorrPct	TrackingErrorPct
Benchmark	100%	100%	100%	100%	0%
MA	88,13%	87,02%	76,24%	75,29%	4,40%
R	86,30%	76,34%	48,34%	72,17%	6,85%
RA	84,47%	73,28%	37,83%	68,89%	7,54%
RAV	84,02%	70,23%	37,15%	68,03%	6,81%
RAVC4_4_4	84,02%	68,70%	40,66%	66,61%	6,55%
RAVC4_3_3	83,56%	66,41%	39,53%	67,13%	6,59%
RAVC5_5_5	85,39%	63,36%	48,47%	63,65%	6,49%
RAVC3_3_3	84,93%	66,41%	39,08%	65,79%	6,70%

Tail events analysis

Strategy	VaR5pct	ES5pct
Benchmark	-0,0421	-0,0542
MA	-0,0372	-0,0491
R	-0,0435	-0,0651
RA	-0,0469	-0,0666
RAV	-0,0364	-0,0522
RAVC4_4_4	-0,0336	-0,0464
RAVC4_3_3	-0,0336	-0,0467
RAVC5_5_5	-0,0329	-0,049
RAVC3_3_3	-0,031	-0,0496



The **most advanced strategies prove to be superior under a wide spectrum of variables considered**.

In particular, **strategies including volatility and correlation factors between assets prove to be more resilient in the face of tail-end events**, of considerable impact and particularly relevant from a long-term perspective.

Looking at the performance generated by the different strategies, the benefits brought by the most advanced strategies are evident. They pay greater attention to the risk component, which plays a decisive role in the final asset allocation, without sacrificing the trend-following component of the classic approach.

In a real operating environment, it can be particularly crucial to have a strategy that can limit volatility, drawdowns and possible negative tail events, while giving up some points of return compared to classic momentum strategies.

This approach, which is more conservative in nature, could be even more prudential through the decision to make a risk-parity allocation among the top assets considered.

$$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{j=1}^{N} \frac{1}{\sigma_j}}$$

In a **risk parity** portfolio, each asset contributes **equally to the overall risk**. The weighting is therefore inversely proportional to its volatility (): riskier assets receive lower weights, while less volatile ones have a higher weight. This approach improves **risk diversification** compared to a simple allocation by capitalization or equal weight.

Modified strategy

The modified approach we propose aims to **verify the effect of introducing volatility and correlation factors on asset selection**. The introduction of these factors reduces the trend-following component typical of momentum strategies based on the lookback of returns alone. The evidence in this regard is already evident from the observation of the results obtained with the replication of the original strategy. Despite this, **we decided to check if this goal could also be achieved with a modified strategy**, albeit not distorted.

The fundamental idea was to accentuate exposure to trend following in relative and absolute momentum strategies.

In order to do **this, it was decided to modify the equally weighted exposure** to top assets proposed in Alpha Architect's replica, to favor a weighting of the weights to be attributed to the assets according to their specific momentum. According to this approach, among the top assets selected**, a greater weight is assigned to those that have recorded a higher momentum in the considered lookback period**.

We believe this accentuates the trend-following nature of the strategy, which is partially limited by the equally weighted allocation of the original replica, which introduces, albeit minimally, a mean reverting component. The attribution of weights according to momentum is therefore done according to this formula:

$$w_{sel} = \frac{MOM_{sel}}{\sum_{i=1}^{7} MOM_{sel,i}}$$

It then specifies how the weight of the single top asset is equal to the momentum that it has recorded on the total momentum for the lookback period of the strategy of the top 7 selected assets.

The modified approach also structures a **new methodology used to measure volatility based on semi-standard deviation**. The choice to use **semi-standard deviation** instead of **classical standard deviation** is justified by the fact that the latter measures volatility by considering **both positive and negative changes** in returns compared to the mean, treating both as risk. However, in an investment environment, investors are generally interested in **limiting losses** and do not perceive positive volatility as an unfavorable element.

Semi-standard deviation, on the other hand, only takes into account **negative deviations** from a given reference value (e.g., mean or zero), thus providing a more consistent measure of the **risk that is actually relevant** from the investor's point of view. In this way, it more accurately represents the **asymmetry of risk** and is particularly useful in the

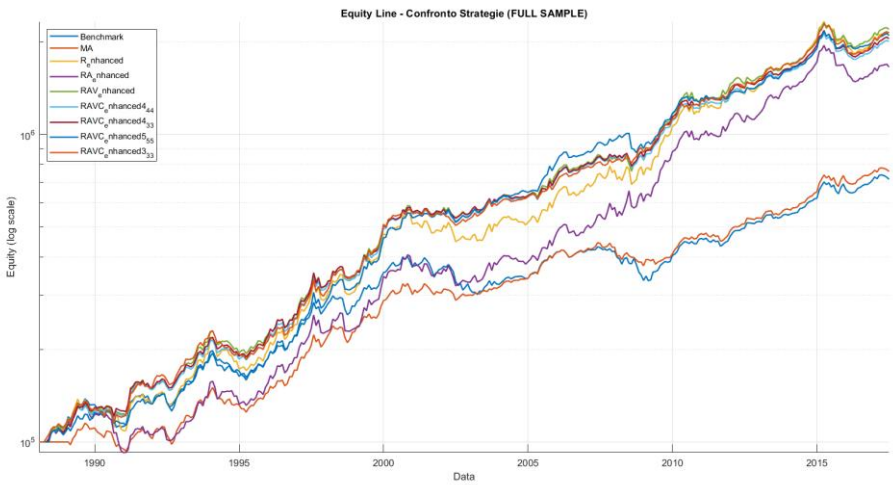
evaluation of investment strategies and in the construction of risk-adjusted performance indicators, such as the **Sortino ratio**.

$$\sigma_{-} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \min(0, r_i - \bar{r})^2}$$

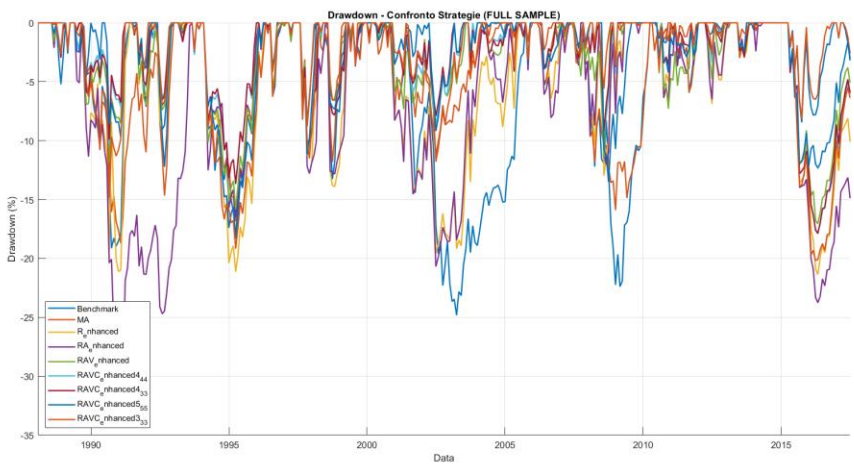
Full sample analysis of the modified strategy

In order to carry out an analysis that is not excessively extensive in the following paper, we limit ourselves to reporting the **results obtained on the entire dataset available**, although the code implemented in MATLAB follows the same structure followed for the main strategy.

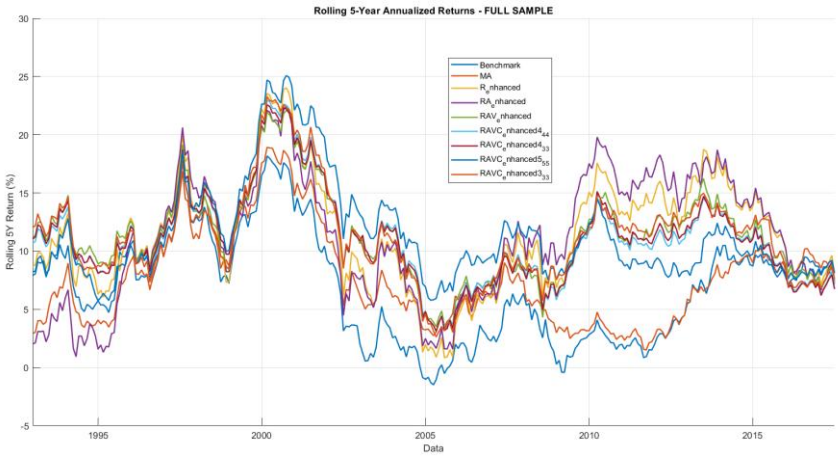
Equity lines



Drawdowns



Rolling 5-year annualized returns



Correlation matrix

	Benchmark	MA	R_enh	RA_enh	RAV_enh	RAVC_enh444	RAVC_enh433	RAVC_enh555	RAVC_enh333
Benchmark	1	0,9	0,7	0,69	0,75	0,74	0,74	0,73	0,73
MA	0,89	1	0,7	0,71	0,74	0,75	0,75	0,76	0,73
R_enh	0,7	0,7	1	0,96	0,93	0,94	0,94	0,88	0,86
RA_enh	0,69	0,7	0,96	1	0,89	0,9	0,9	0,88	0,82
RAV_enh	0,75	0,7	0,93	0,89	1	0,97	0,97	0,9	0,9
RAVC_enh444	0,74	0,8	0,94	0,9	0,97	1	0,92	0,92	0,92
RAVC_enh433	0,74	0,8	0,94	0,9	0,97	0,92	1	0,92	0,92
RAVC_enh555	0,73	0,8	0,88	0,88	0,9	0,92	0,92	1	0,87
RAVC_enh333	0,73	0,7	0,86	0,82	0,9	0,92	0,92	0,87	1

Annual returns of individual strategies

Anno	Benchmark	MA	R_enhanced	RA_enhanced	RAV_enhanced	RAVC_enhanced4_4_4	RAVC_enhanced4_3_3	RAVC_enhanced5_5_5	RAVC_enhanced3_3_3
1988	13,39%	0,00%	13,28%	11,05%	13,23%	10,48%	10,12%	8,08%	14,42%
1989	16,65%	11,11%	12,48%	10,85%	13,30%	14,44%	17,74%	15,41%	12,63%
1990	-14,97%	-14,63%	-14,59%	-24,98%	-3,74%	-3,23%	-2,62%	-7,73%	-6,06%
1991	18,15%	8,04%	23,18%	14,25%	19,97%	20,84%	18,64%	15,39%	25,06%
1992	11,67%	12,74%	13,34%	4,77%	14,51%	12,30%	12,72%	10,15%	16,49%
1993	27,44%	26,17%	41,23%	41,23%	33,43%	28,27%	28,94%	34,25%	29,35%
1994	-10,05%	-9,47%	-19,68%	-14,96%	-13,79%	-12,74%	-12,70%	-16,67%	-15,16%
1995	9,76%	9,77%	15,81%	17,07%	14,03%	16,28%	16,82%	14,62%	10,13%
1996	18,33%	17,98%	20,80%	21,06%	16,78%	17,72%	17,03%	19,26%	18,18%
1997	23,98%	25,90%	21,16%	21,16%	23,85%	24,09%	24,26%	27,63%	21,65%
1998	0,76%	6,11%	5,62%	2,67%	7,14%	7,88%	7,93%	11,92%	12,90%
1999	28,65%	26,16%	54,26%	47,56%	46,37%	44,95%	44,97%	44,16%	47,42%
2000	4,10%	4,79%	7,36%	7,82%	9,57%	11,65%	10,68%	16,29%	8,91%
2001	-1,00%	1,08%	-5,33%	-5,47%	0,52%	-0,88%	-0,06%	2,58%	-0,77%
2002	-14,63%	-1,51%	-5,99%	-6,01%	-0,25%	-0,04%	-0,12%	-0,67%	-3,22%
2003	4,13%	6,07%	13,89%	17,23%	11,68%	12,39%	11,27%	13,17%	12,19%
2004	4,60%	6,84%	0,03%	-0,63%	2,40%	2,16%	1,70%	4,94%	4,82%
2005	21,92%	21,58%	23,07%	24,62%	18,16%	16,03%	15,94%	26,18%	11,90%
2006	0,53%	1,43%	6,26%	3,51%	7,08%	8,98%	9,19%	6,51%	11,00%
2007	-0,04%	2,27%	12,85%	16,44%	4,80%	6,96%	7,09%	10,21%	5,06%
2008	-19,61%	-12,94%	-3,44%	4,51%	0,34%	-2,85%	-1,28%	-6,23%	6,72%
2009	20,49%	9,36%	44,12%	45,44%	38,17%	34,65%	34,86%	30,17%	28,62%
2010	13,64%	16,31%	15,88%	13,67%	12,05%	13,61%	13,46%	12,09%	15,03%
2011	1,61%	2,26%	11,68%	11,25%	11,74%	6,15%	6,67%	3,70%	8,91%
2012	11,75%	9,94%	4,36%	4,72%	5,68%	9,24%	8,48%	6,39%	7,83%
2013	4,92%	6,83%	18,82%	23,26%	10,07%	9,62%	9,25%	11,64%	11,83%
2014	13,42%	13,81%	20,37%	19,67%	18,67%	19,18%	19,30%	17,47%	18,99%
2015	6,82%	8,80%	-1,85%	-5,23%	0,90%	-2,43%	-2,56%	0,12%	-3,64%
2016	9,78%	8,08%	3,15%	0,97%	5,33%	4,08%	4,69%	8,09%	5,73%
2017	-1,70%	-0,88%	0,60%	0,75%	2,46%	3,36%	2,95%	2,16%	3,94%

Summary statistics

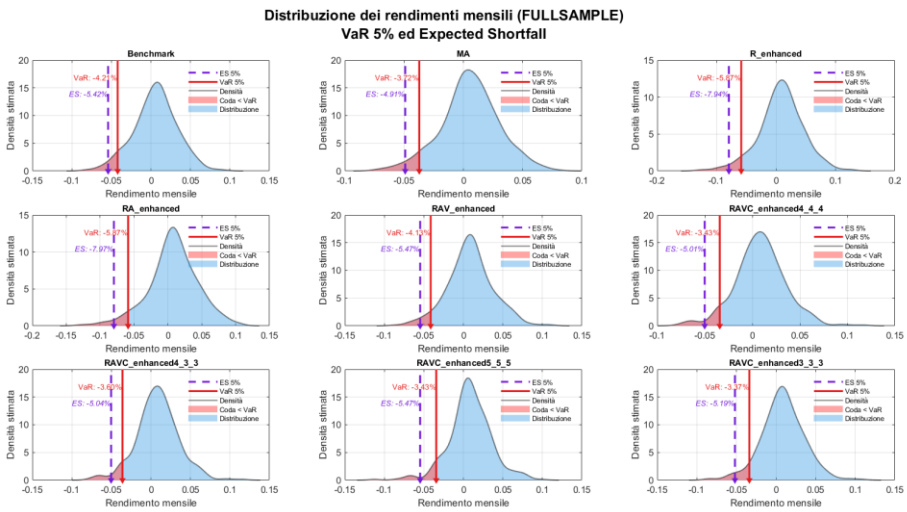
Strategy	CAGR	Volatility	Sharpe	Sortino	SortinoMod	Calmar	Validity	Omega
Benchmark	0,0691	0,0948	0,2496	0,2848	0,2844	0,2786	0,1119	0
MA	0,0713	0,0849	0,3023	0,3525	0,338	0,3808	0,1862	0,7767
R_enhanced	0,1089	0,1303	0,462	0,5983	0,3219	0,5098	0,311	1,5548
RA_enhanced	0,1002	0,1269	0,4119	0,5045	0,2922	0,3214	0,2568	0,6130
RAV_enhanced	0,1107	0,1005	0,615	0,8867	0,4582	0,6493	0,6462	3,5339
RAVC_enhanced4_4_4	0,1073	0,0927	0,6334	0,8944	0,4847	0,6023	0,6231	3,3896
RAVC_enhanced4_3_3	0,1081	0,0929	0,6404	0,9039	0,4853	0,6045	0,72	3,6883
RAVC_enhanced5_5_5	0,1091	0,0938	0,643	0,8082	0,436	0,6023	0,7653	3,6420
RAVC_enhanced3_3_3	0,1097	0,0929	0,6559	0,9141	0,4815	0,5436	0,6771	3,3282

Correlation analysis

Strategy	UpPct	DownPct	NegCorrPct	PosCorrPct	TrackingErrorPct
Benchmark	100,00%	100,00%	100,00%	100,00%	0,00%
MA	88,13%	87,02%	76,24%	75,29%	4,40%
R_enhanced	81,74%	71,76%	33,56%	64,53%	9,28%
RA_enhanced	81,74%	67,94%	38,61%	57,99%	9,24%
RAV_enhanced	85,39%	70,23%	31,83%	71,84%	6,95%
RAVC_enhanced4_4_4	84,47%	68,70%	34,95%	69,94%	6,75%
RAVC_enhanced4_3_3	84,02%	66,41%	34,25%	70,50%	6,78%
RAVC_enhanced5_5_5	86,30%	62,60%	38,92%	66,15%	6,91%
RAVC_enhanced3_3_3	85,39%	66,41%	34,82%	66,21%	6,94%

Tail events analysis

Strategy	VaR5pct	ES5pct
Benchmark	-0,0421	-0,0542
MA	-0,0372	-0,0491
R_enhanced	-0,0587	-0,0794
RA_enhanced	-0,0587	-0,0797
RAV_enhanced	-0,0413	-0,0547
RAVC_enhanced4_4_4	-0,0343	-0,0501
RAVC_enhanced4_3_3	-0,036	-0,0544
RAVC_enhanced5_5_5	-0,0343	-0,0547
RAVC_enhanced3_3_3	-0,0337	-0,0519



Conclusions

In both experiments conducted, an improvement in performance was observed **following the introduction of volatility and correlation factors**.

In the original strategy, the more traditional approaches among the proposed strategies have recorded higher returns, but lower overall return-risk efficiency than more advanced strategies.

In the pure replication experiment, the RAVC 444 strategy demonstrated superior performance under the broad spectrum of indicators taken into consideration. In the hierarchy of our evaluations**, we have chosen to favor the OPR**, as we believe it is a particularly suitable measure for accurately measuring the performance of each portfolio compared to its Benchmark.

In the modified strategy, the stated objective was to **accentuate the trend-following component of classic momentum strategies**, in order to verify the effectiveness of the additional volatility and correlation filters introduced in the more advanced strategies. **In this scenario, the beneficial nature of these factors emerged even more evidently as tools capable of dampening the trend-following component of the classic versions**. The breadth of the time series available allows us to analyze various cycles of the financial markets, such as to be able to include both trend following phases and mean reverting phases.

In the Framework considered, the worst strategy among those of the FAA turned out to be RA enhanced. This strategy, by its nature, excludes assets with negative returns in the lookback window. The increased exposure to trend following sought has shown an inability to take advantage of the positive rebounds of assets with negative returns.

The correction of this single factor through volatility and correlation factors has allowed us to further confirm what we have already seen in the faithful replication of the strategy. **In the analysis carried out with the modified strategies, the strategy with the highest OPR was RAVC 433 enhanced**, a strategy strongly correlated by its nature to RAVC 444 enhanced, whose alias is the winning strategy in the replication experiment.

In general, advanced strategies have shown a remarkable ability to reduce the volatility of R and RA strategies, which are now characterized by a greater concentration in terms of asset allocation, which is linked to greater volatility than the original strategies with an Equally Weighted weighting on the selected top assets. **They also generally experienced a higher CAGR than the traditional approach**.

In this perspective, **the use of semi-standard deviation as a risk indicator seems to have been particularly effective for the most advanced strategies,** in particular **RAV enhanced was the most profitable strategy in its modified version**.

Ultimately, we believe that the experiment can be expanded by using more challenging benchmarks, such as a simple Buy and Hold allocation in a global equity index and through the introduction of transaction costs as a tool capable of assessing the actual operational viability of the strategies examined.